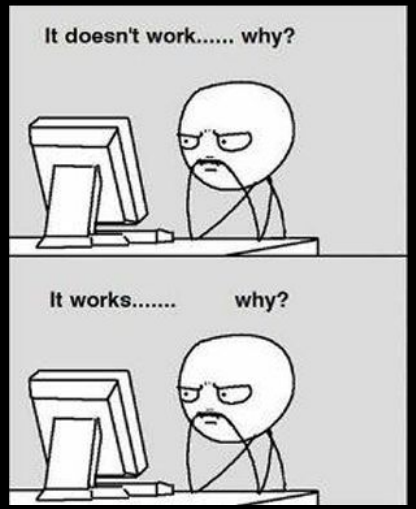
# PROMPTING IN LLMs

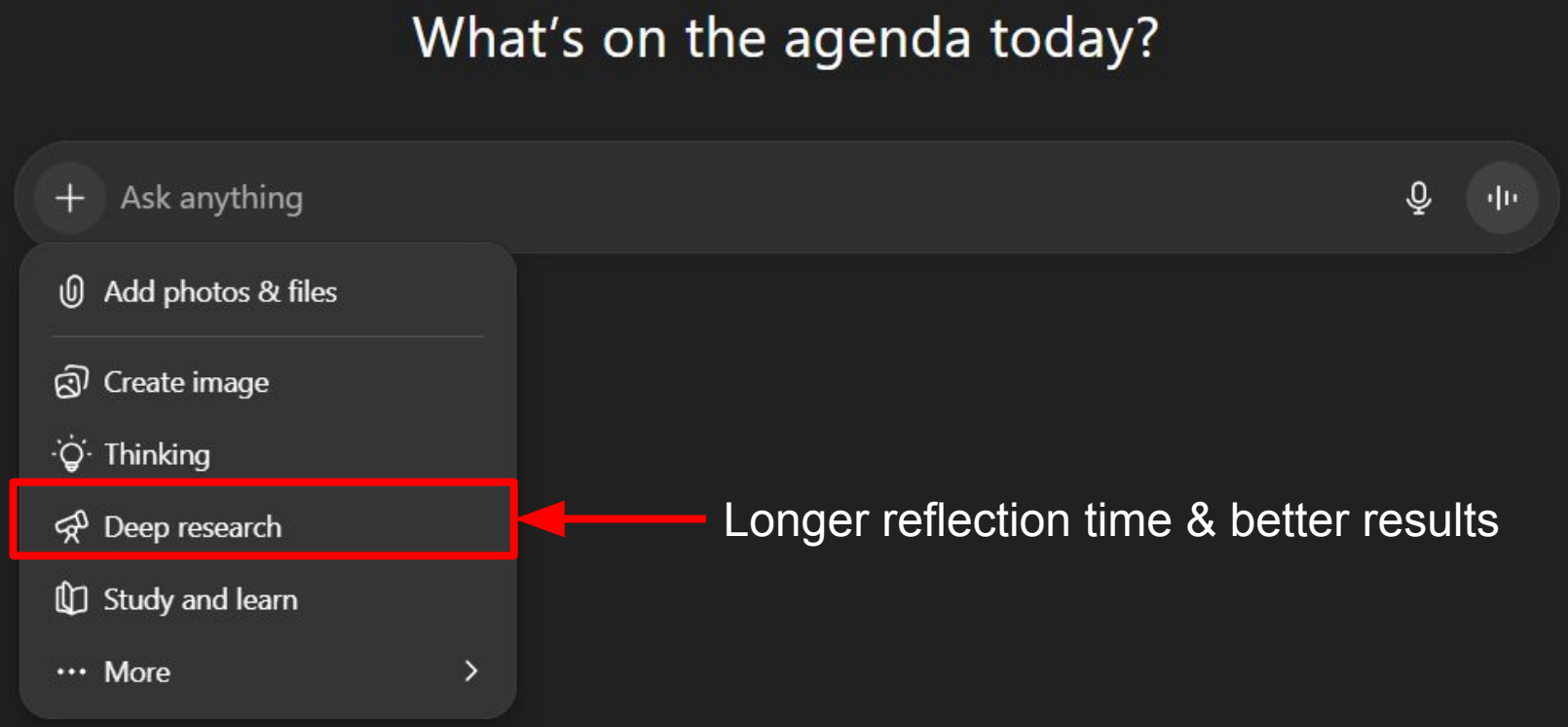
## Enhance your code with AI




Python and Data Tools for Non-Programmers - Aadim Nyaichyai

# CHOICES?? ... A lot of choices.

- ChatGPT
- Copilot (Microsoft)
- Gemini (Google)
- Also, Deepseek, Claude…




Longer reflection time & better results

Help me get better with Python.

Prompt

Absolutely — I can help you level up your Python skills.

Here's a **structured, practical path** we can follow, and you can tell me where you want to start.

Output

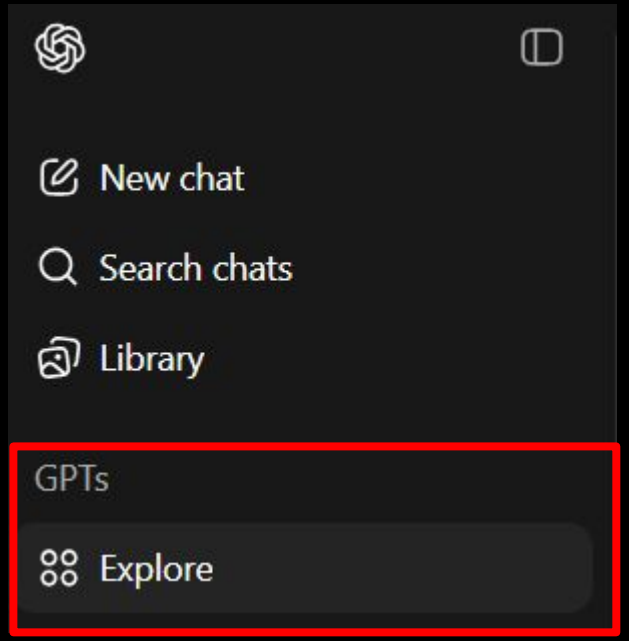
---

## 🚀 Your Python Skill-Up Plan

### 1. Strengthen the Foundation (if needed)

We can review or practice:

+ Ask anything


New chat
Search chats
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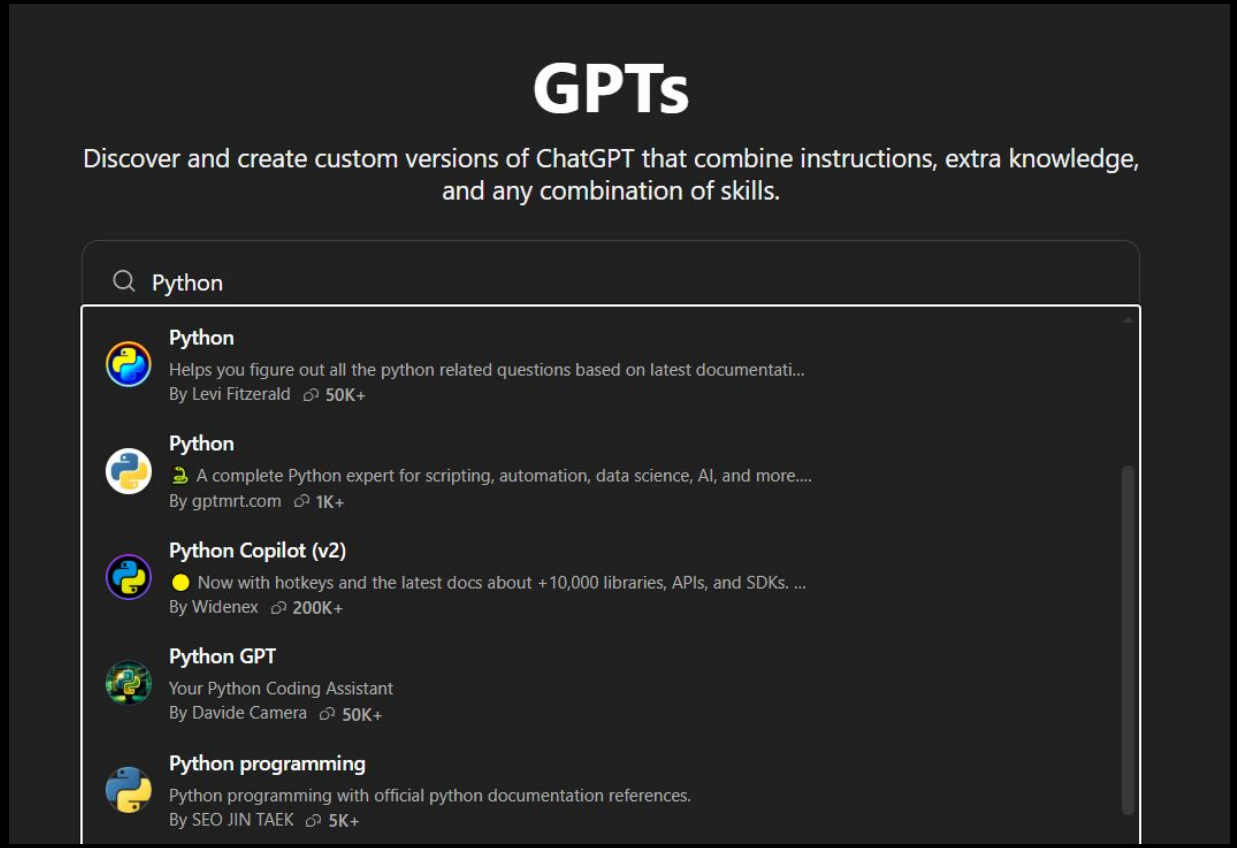
GPTs
Discover and create custom versions of ChatGPT that combine instructions, extra knowledge, and any combination of skills.
Python
Python
Helps you figure out all the python related questions based on latest documentati...
By Levi Fitzerald 50K+
Python
A complete Python expert for scripting, automation, data science, AI, and more....
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Now with hotkeys and the latest docs about +10,000 libraries, APIs, and SDKs. ...
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Python programming
Python programming with official python documentation references.
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# Writing code from scratch

# BE SPECIFIC

Too vague

Write a function to check for palindromes

Specific

Write a python function named is_palindrome() that takes a parameter 'n', which can be a string, numerical or alphanumerical. The function should:
- Account for case-sensitivity.
- Check for any spaces & clean it.
- Checks for empty strings.
- Print an output that states if the input is a palindrome or not.

# PSEUDOCODES

- Pseudocode is not a real programming language.
- Pseudocode is an informal way of representing the steps of a program.
- Breaks down complex problems into smaller, more manageable steps.

**Step-by-Step Instructions:**

1. **Create a list named `features` containing some floating-point numbers representing some measurements (e.g., lengths, widths).**
2. **Create a dictionary named `data_point` where keys are strings describing the features and values are the corresponding items from the `features` list.**
3. **Print the dictionary to verify its contents.**

**Example Input: features = [4.5, 3.2, 1.5] Expected Output: data_point = {'length': 4.5, 'width': 3.2, 'depth': 1.5}**

# Clear Input = Clear Output

```
A program to calculate the fctorial of a number.

1. Ask the user for a number 'n'.
2. Check if the input is a number.
3. Initialize a variable 'factorial' to 1
4. Calculate the factorial of 'n'
5. Print the value of 'factorial'.
```

```
A program to calculate the fctorial of a number.

1. Ask the user for a number 'n'.
2. Check if the input is a positive number or not.
    - If 'n' is not a positive integer:
        - Ask for a positive integer.
3. Initialize a variable 'factorial' to 1
4. If 'n' is 0 or 1:
    - Set 'factorial' to 1
   Else:
    - For i from 1 to 'n':
        - Multiply 'factorial' by 'i'.
5. Print the value of 'factorial'.
```

# SET BOUNDARIES

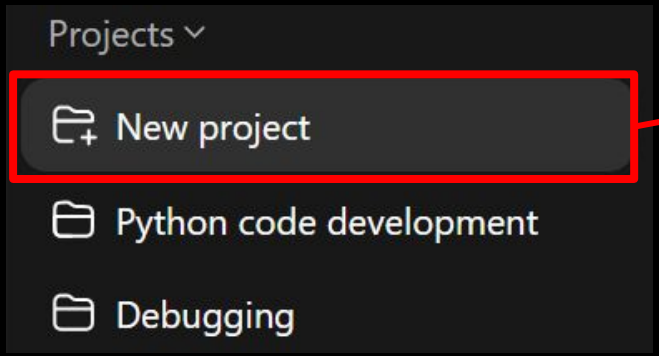
- Use only built-in functions
- No advanced concepts (like recursions)

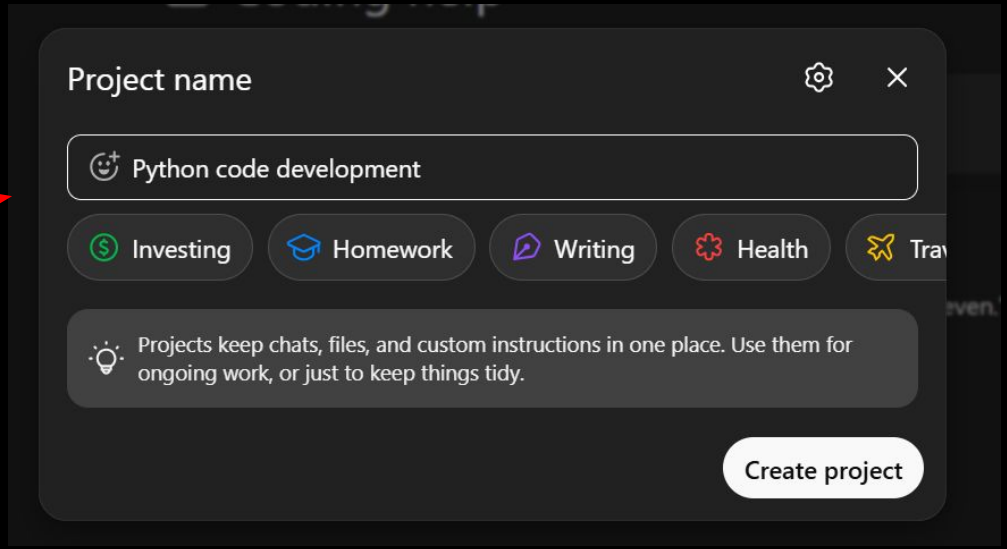
only use the following modules:
- Numpy
- SciPy
- Pandas
- Matplotlib

Fortunately, ChatGPT iterative. So, you don’t need to set boundaries everytime you want to write a new code.

However, for better management you can also create a dedicated ChatGPT project and set the boundaries as instructions.

# PROJECTS


Projects
New project
Python code development
Debugging


Project name
Python code development
Investing
Homework
Writing
Health
Projects keep chats, files, and custom instructions in one place. Use them for ongoing work, or just to keep things tidy.
Create project

Kuenneth

ChatGPT

Get Plus

Share

Edit project

Edit instructions

Delete project

# Python code development

New chat in Python code development

**Python palindrome function**
Write a python function named is_palindrome() that takes a parameter 'n', which can be ... Nov 17

**Even or odd program**
write a program to check if any number is even or odd Nov 2

Python code development

Instructions

**How can ChatGPT best help you with this project?**
You can ask ChatGPT to focus on certain topics, or ask it to use a certain tone or format for responses.

1. Whenever I ask for some code, please provide comments to go along with it. For functions, add a docstring. Docstrings should follow Google's documentation style.
2. Remember I am just a beginner.
3. Please only use built-in-functions, functions from matplotlib, NumPy & pandas.

Cancel Save

Instructions will only be followed in the chats within the projects.

# Ask chatGPT to document for you

1. Whenever I ask for some code, please provide comments to go along with it. For functions, add a docstring. Docstrings should follow Google's documentation style.

- Docstrings for functions.
- # comments for code snippets.

python Copy code

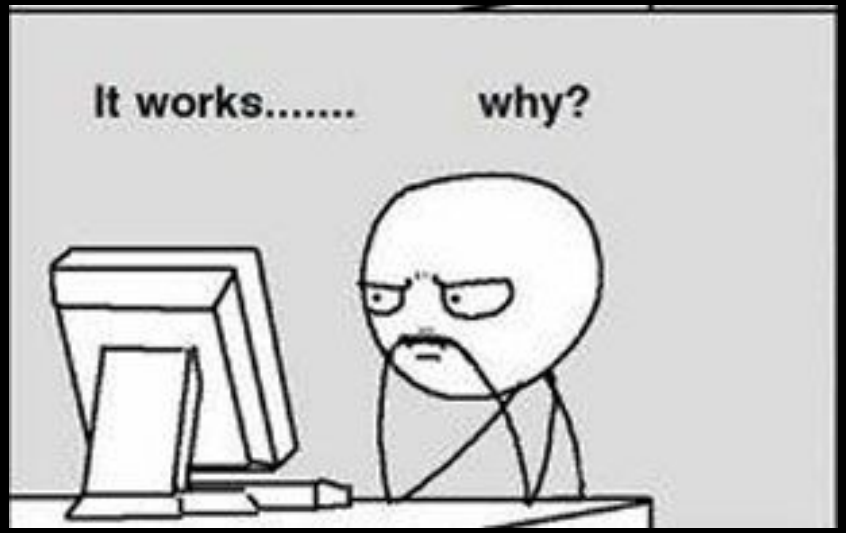
```python
def is_palindrome(n):
    """Check if a given input is a palindrome.

    Args:
        n: A string, number, or alphanumeric value to be checked.

    Returns:
        None. Prints whether the input is a palindrome or not.
    """
    # Convert the input to a string so we can process it
    text = str(n)

    # Clean spaces from the string
    cleaned = text.replace(" ", "")
```

# Okay now your code works!!



# Explaining code

- Assign a specific role (eg… You are a python tutor..)
- Set the audience level (eg. explain to a 10 year old..)
- Instruct to use familiar concepts and avoid technical jargon.

Act as a seasoned python programmer with 20 years of experience. I am a 10-year-old with limited vocabulary, so avoid technical jargons and use familiar concepts. I will provide you a code and you have to explain:
1. how does it work step-by-step?
2. Explain the logic behind the approach.
3. Elaborate the syntax of any function used. (How many parameters it takes, what it does, what it returns)
4. Are there any potential issues or limitations with this approach?

# Coding help

## Instructions

**How can ChatGPT best help you with this project?**
You can ask ChatGPT to focus on certain topics, or ask it to use a certain tone or format for responses.

The purpose of this project is to clarify what a certain piece of code is meant to do. Act as a seasoned python programmer with 20 years of experience. I am a 10-year-old with limited vocabulary, so avoid technical jargons and use familiar concepts. I will just provide you a code and you have to explain:
1. how does it work step-by-step?
2. Explain any logic behind the approach.
3. Elaborate the syntax of any function used. (How many parameters it takes, what it does, what it returns)
4. Are there any potential issues or limitations with this approach?

Cancel Save

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print(f"{num} is even.")
else:
    print(f"{num} is odd.")
```

Here's your explanation in a **simple, kid-friendly way**, just like you asked 😊

---

## ⭐ 1. How does this code work step-by-step?

### Step 1: Ask the user for a number

python Copy code

```python
num = int(input("Enter a number: "))
```

- `input("Enter a number: ")` shows a message on the screen asking the user to type a number.
- Whatever the user types is read as *text* (a string).

## ⭐ 2. What is the logic behind this?

The idea is super simple:

- **Even numbers** divide by 2 with *no leftovers*.
- **Odd numbers** always have *1 leftover* when divided by 2.

Example:

- 10 ÷ 2 = 5 → remainder 0 → even
- 7 ÷ 2 = 3 → remainder 1 → odd

Python uses `% 2` to check the remainder.

---

## ⭐ 3. Function / syntax explanations

`input(prompt)`

- **Takes:** 1 parameter → a message to show the user (like `"Enter a number: "`).
- **Does:** Waits for the user to type something.
- **Returns:** A string.

`int(value)`

- **Takes:** 1 value (usually a string that looks like a number).
- **Does:** Converts that value to an integer number.

# Debugging

# Types of bugs in Python

Compilation Errors:

- When the code doesn't follow the syntax rules of the programming language.

```
In [1]:
num = float(input("Enter a number: "))
if num > 0:
    print("Positive number")
elif num == 0
    print("Zero")
else:
    print("Negative number")

  File "C:\Users\aadim\AppData\Local\Temp/ipykernel_700
8/2923611721.py", line 4
    elif num == 0
                 ^
SyntaxError: invalid syntax
```

Logical errors:

- Code runs without any error messages, but does not produce desired output.

```
In [1]:
'''
Checks if the number is even or odd.
'''
x = 24
if x % 2 == 0:
    print("Odd")
else:
    print("Even")

Odd
```

# Compilation errors

- Include the exact error message + code.
- Ask for simple explanations.
- Request fixes.

```
In [2]:
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)

print(factorial(a))
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
~\AppData\Local\Temp/ipykernel_9020/2340683917.py in <module>
      5         return n * factorial(n - 1)
      6
----> 7 print(factorial(a))

NameError: name 'a' is not defined
```

Kuenneth Group

Debugging

Instructions

**How can ChatGPT best help you with this project?**

You can ask ChatGPT to focus on certain topics, or ask it to use a certain tone or format for responses.

Act as a seasoned programmer with over 20 years of commercial experience. Analyze the provided piece of code that is causing a specific error. Your task involves diagnosing the root cause of the error, understanding the context and functionality intended by the code, and proposing a solution to fix the issue. Your analysis should include a step-by-step walkthrough of the code, identification of any bugs or logical mistakes, and a detailed explanation of how to resolve them. Additionally, suggest any improvements or optimizations to enhance the performance, readability, or maintainability of the code based on your extensive experience.

Cancel Save

Don't forget to be specific.

Hey I got this error message:

```
----> 7 print(factorial(a))

NameError: name 'a' is not defined
```

Here's my code:

```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)

print(factorial(a))
```

## ✅ How to Fix It

You need to *define* `a` before using it.

Here are two correct options:

---

### ✅ Option 1: Assign a value to `a`

python Copy code

```python
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n - 1)

a = 5   # or any number you want
print(factorial(a))
```

# Logical error

- Explain what the code is intended to do. (i.e. Pseudocodes)
- Break down the code into smaller, complete units eg: if there is a function, loops, conditions.

Hey, there seems to be something wrong with my code and I cannot find what it is. It's supposed to tell if a number is even or odd but it's not working as intended.

```
x = 24
if x % 2 == 0:
    print("Odd")
else:
    print("Even")
```

Here's a clean breakdown of what's going wrong — it's a simple but very common logical bug.

---

### 🔍 Diagnosis

You want the program to print **"Even"** if the number is even, and **"Odd"** otherwise.

But your current code does the opposite:

```python
if x % 2 == 0:
    print("Odd")   # ← Wrong label
else:
    print("Even")  # ← Wrong label
```

- `x % 2 == 0` means **x is even**, but you are printing `"Odd"`.
- The `else` branch is printing `"Even"` for odd numbers.

# Emotional Prompts- do they work?

Can you write a python program to check for duplicates in a dataframe.

Emotional stake

Hey, I'm in big trouble right now. I am really stressed out with my course work and I don't think I can do this on my own. I really need your help today. If I don't get this done properly I will fail this class and waste a year of my life. I need you to write a python program to check for duplicates in a dataframe. I really need you to get this right and I trust you with all my heart. Thank you!

## ✅ Full Example Program

Here is a complete script you can run:

python Copy code

```python
import pandas as pd

def check_duplicates(df, subset=None):
    """
    Returns the duplicate rows in a DataFrame.

    subset: list of column names to check for duplicates (optional)
    """
    duplicates = df[df.duplicated(subset=subset, keep=False)]

    if duplicates.empty:
        print("No duplicates found.")
    else:
        print("Duplicate rows:")
        print(duplicates)

# Example usage
if __name__ == "__main__":
    data = {
        "id": [1, 2, 3, 4, 3],
        "name": ["A", "B", "C", "D", "C"],
        "value": [10, 20, 30, 40, 30]
    }

    df = pd.DataFrame(data)
    check_duplicates(df, subset=["id"])
```

## 🔷 Full Program

python Copy code

```python
import pandas as pd

def check_duplicates(df, subset=None):
    """
    Check for duplicate rows in a pandas DataFrame.

    Args:
        df (pd.DataFrame): The DataFrame to check.
        subset (list or str, optional): Column name(s) to check duplicates for.
                                        If None, check duplicates based on all co
    """
    print("\n=== Checking for duplicates ===")

    # Identify duplicates
    duplicates = df[df.duplicated(subset=subset, keep=False)]

    if duplicates.empty:
        if subset:
            print(f"No duplicates found based on column(s): {subset}")
        else:
            print("No duplicate rows found in the DataFrame.")
    else:
        if subset:
            print(f"Duplicates found based on column(s): {subset}")
        else:
            print("Duplicate rows found in the DataFrame:")
        print(duplicates)

    return duplicates
```

# Summary

- Be specific with your prompts!
- Set boundaries.
- Assign a role.
- Set the audience level.
- Break down the task.
- Ask for documentation.

You are an experienced programmer in Python. You will be guiding a high school student to write a Python function to return the difference between the largest and smallest numbers that can be formed with its digits. The function should:
1. Be named 'rearranged_difference' that takes an integer 'n' as input.
2. Rearrange the digits to form the largest and smallest possible numbers.
3. Handle negative numbers by working with their absolute value.
4. Include a docstring & comments with proper explanations.
5. Explain the syntax & functionality of any function used in the code. Your explanation should be comprehensive enough to cater to novice programmers. Break down the code's functionality, explain its logic.

python

Copy code

```python
def rearranged_difference(n):
    """
    Return the difference between the largest and smallest numbers
    that can be formed by rearranging the digits of the integer n.

    How it works:
    1. Convert n to its absolute value so negative signs are ignored.
    2. Convert the number into a string to access each digit.
    3. Turn the string into a list of digits.
    4. Sort the list in descending order to form the largest number.
    5. Sort the list in ascending order to form the smallest number.
    6. Convert both lists back into integers.
    7. Return the difference between them.
    """
    # Step 1: Work with the absolute value to ignore negative signs.
    n = abs(n)

    # Step 2: Convert the number to a string to access each digit.
    digit_string = str(n)

    # Step 3: Convert the string of digits into a list.
    digits = list(digit_string)

    # Step 4: Sort digits in descending order to form the largest number.
    largest_num = int(''.join(sorted(digits, reverse=True)))

    # Step 5: Sort digits in ascending order to form the smallest number.
    smallest_num = int(''.join(sorted(digits)))

    # Step 6: Return the difference.
    return largest_num - smallest_num
```

## 🧩 Explanation of Python Functions Used

`abs(n)`

- Returns the absolute (positive) value of a number.
- Example: `abs(-15)` → `15`

`str(n)`

- Converts a number into its string version.
- Example: `str(123)` → `"123"`

`list("123")`

- Turns a string into a list of its characters.
- Example: `['1', '2', '3']`

`sorted(iterable)`

- Returns a new list where the items are in ascending order.
- Example: `sorted([3, 1, 2])` → `[1, 2, 3]`

## 🧠 Breakdown of the Logic (Step-by-Step)

### Step 1 — Take the absolute value

Negative numbers include a minus sign `'-'`, which is **not a digit**, so we remove it using `abs()`.

### Step 2 — Convert the number into a string

You cannot loop through an integer directly.
Turning it into a string allows you to access each digit individually.

Example:
`str(572)` → `"572"`

### Step 3 — Create a list of digits

Using `list()` turns the string into a list of characters:

# References


- ChatGPT Prompt Engineering for Developers - DeepLearning.AI
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- ChatGPT Prompting Cheat Sheet: How to Write Better Prompts with GPT-4.1 (Beginner-Friendly Guide) - Bruhith Indiresh